

Problem 2: Exploring Edge Impulse

1. Learning Blocks available

- Classification: Categorizes input data. Example: classifying different movements or audio commands.
- Transfer Learning (Images): Fine-tunes a pretrained image-classification model. Example: classifying images of different objects.
- Regression: Predicts continuous numerical values. Example: predicting temperature from sensor data.
- Visual Anomaly Detection – FOMO-AD: Detects unusual regions in images without requiring anomalous training samples. Example: identifying defects on manufactured products.
- Anomaly Detection: Finds unusual samples or outliers. Example: detecting abnormal vibration in a machine.

2. Classifier layers

- Dense: A fully connected layer used to learn relationships between features.
- 1D Convolution / Pooling: Extracts and reduces patterns in one-dimensional data such as audio or sensor signals.
- 2D Convolution / Pooling: Extracts and reduces spatial patterns in images or spectrograms.
- Reshape: Changes the dimensions of data so it can enter another layer.
- Flatten: Converts multidimensional data into a one-dimensional vector.
- Dropout: Randomly disables connections during training to reduce overfitting.

3. Model compression options

- Quantized (int8): Converts the model to 8-bit integers to reduce memory usage and model size.
- EON Compiler: Optimizes the model for lower RAM and ROM usage.

The unoptimized float32 option is also available, but it is not a compression method.

4. Synthetic-data modalities

- Images:
 - Black Forest Labs Flux-Pro Synthetic Image Generator
 - OpenAI GPT-Image-2 Synthetic Image Generator
 - NVIDIA Omniverse
- Audio/voice:
 - OpenAI Whisper Synthetic Voice Generator

Synthetic data helps increase dataset size, balance classes, create rare examples, reduce data-collection costs, and improve model robustness.